The AI talent market: A Statistical Analysis of Salary Gaps and Work Mode Trends

Overview

This project investigates the AI job market by analyzing the relationships between key variables. Specifically, it examines how salary correlates with company size and work modes, and whether company sizes and work modes are related to each other. A series of tests and complementary analyses were performed in order to examine these relationships. Primarily, the correlation between salary and considered extremely flexible and inflexible work modes (remote and on-site, respectively) correlation were examine, in which a non-significant relationship between them was found [1]. On the other hand, a strict correlation between company sizes and salaries was measured. In addition, the relationship examination between company sizes and work modes (this time considering all types of flexible and non-flexible modes) showed a strong correlation among them, forming a U-shaped pattern with startups (left peak) making extensive use of this work mode, mid-sized (bottom) companies restricting it as the scale, and large enterprises (right peak) reintroducing it using global infrastructure [3].

1. The test was conducted in order to evaluate the pay gap in salaries across extremely flexible and non-flexible work arrangements cultures.
2. Due to the imbalanced sample size across company tiers, a non-parametric Kruskal-Wallis test was applied, followed by a Mann-Whitney U post hoc test to draw specific salary gaps.
3. Although the test would prove the correlation between the variables, the relationship was not emphasized. Posteriorly, a heatmap was plotted in order to visualize this interaction.

Dataset

The project dataset contains 71,913 records of AI jobs between 2020 and 2026. Some cleaning approaches were applied, considering the tests that could be applied, in order to avoid biased results. The strategy consisted of verifying the test variables, finding outliers, and using an interquartile range technique (IQR) to drop values falling beyond this range. In addition, another

aspect considered was the data density, specifically those related to categorical variables, since a significant spread difference between them could also generate a biased result. These features were used to determine the most suitable test for each hypothesis in the generation of a valuable conclusion.

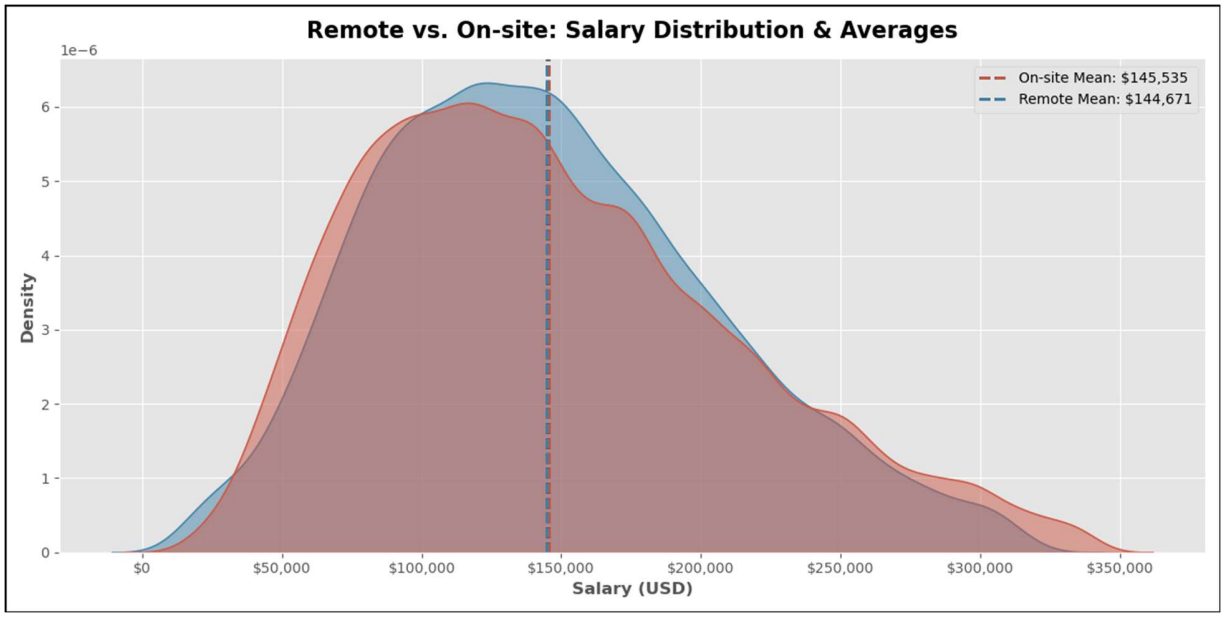
Findings

Experiment 1: Is there a salary gap between on-site and remote modes?

The first experiment sought to determine whether there is a salary compensation gap between remote workers and those who work in traditional offices. Salary (USD) and the peak of flexible and non-flexible work arrangements (remote and on-site, respectively) variables were set to perform the test. The test definitions are the following:

- **Null Hypothesis (H_0):** $\mu_{\text{remote}} = \mu_{\text{onsite}}$ (mean salaries are identical between the work modes)
- **Alternative Hypothesis (H_1):** $\mu_{\text{remote}} \neq \mu_{\text{onsite}}$ (mean salaries are not identical between the work modes)
- **Alpha (α):** 0.05

To guarantee a suitable test selection, Levene's test was run to verify the data homogeneity across on-site and remote work modes; consequently, the measured result ($p = 1.37 \times 10^{-34}$) was less than the threshold (0.05), leading to a Welch's t-test selection to evaluate whether there is a significant salary difference between on-site and remote modes. The value found ($p = 0.11$) indicates a non-significant salary difference between the analyzed work arrangements. However, the presented result is an overall analysis between these work arrangements; therefore, there is no evidence that suggests a pay gap exists considering other attributes through just the applied examination. Posteriorly, additional measurement methods were conducted in order to investigate potential salary differences and their causes.



Graphic 1: Salary distribution and average across remote and on-site work modes.

The distributions between on-site and remote arrangements were displayed through a histogram. A similar salary density is observed across these modes, where the salary mean lines evidence this behavior, as well as the almost symmetrical distribution shape formed by the two modes. Therefore, the obtained result showed that there is no strict difference in the salary composition comparison across the peak of flexible and non-flexible work arrangements, so that they are not individually the cause of a possible pay gap. Accordingly, other dataset aspects were evaluated to better comprehend the AI job market composition.

Experiment 2: Is there a significant salary difference across company sizes?

To continue examining salary composition behavior, the second experiment analyses its distribution across different company sizes, to determine whether they have distinct effects on the salary average. Salary (USD) and company size (large, medium, and small) variables were compared to evaluate this correlation. The test definitions are the following:

- **Null Hypothesis (H_0):** $\mu_S = \mu_M = \mu_L$ (There is no influence of company size on the salary mean).
- **Alternative Hypothesis (H_1):** At least one company size has a different population mean salary.

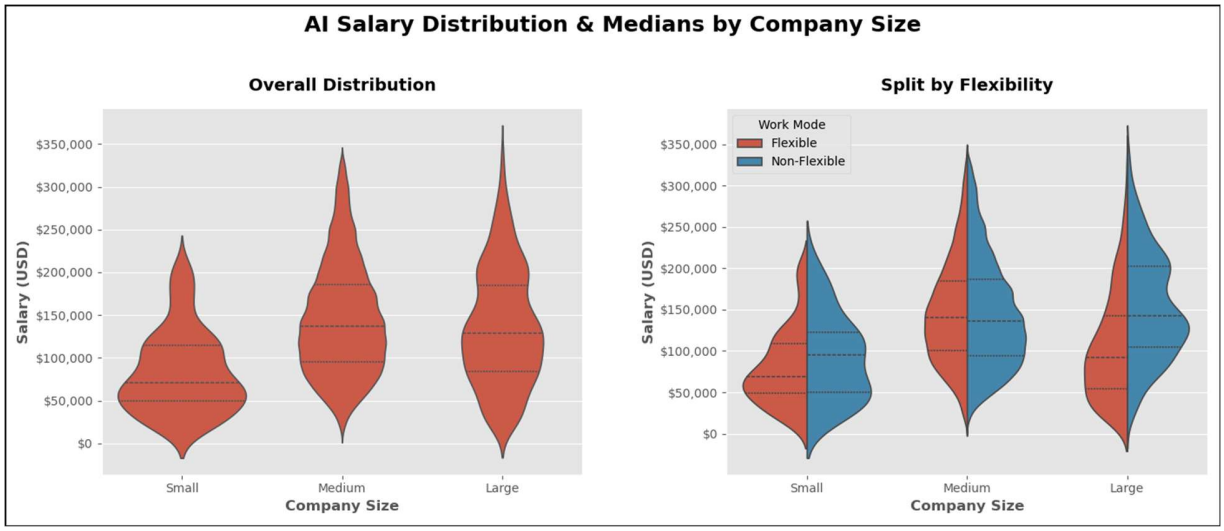
- **Alpha (α):** 0.05

Initially, a data spread check was conducted to determine whether there is a similar data density across company sizes. The result showed a considerable difference in their distribution, with small companies accounting for 215 (0.30%) registers, medium companies accounting for 70,110 (97.49%), and large companies accounting for 1,558 (2.21%). Once the measured data distribution presented a high dissimilarity, a Kruskal-Wallis test was selected to analyze the association between company size and salary mean. Subsequently, an IQR technique was performed to drop outlier salaries and avoid biased results. The test result ($p = 3.44 \times 10^{-50}$) indicates an association between salary and company size, providing evidence that there is a salary difference across sizes.

Company Size	Count	Density (%)
Small	215	0.30%
Medium	70,110	97.49%
Large	1,558	2.21%
Total	71,913	100%

Table 1: Company size distribution before interquartile technique.

Subsequently, a Mann-Whitney U post hoc test was performed evaluating the correlation between all sizes and showed a strong correlation between medium and large companies, as well as between other sizes: small versus medium-sized companies ($p = 2.570 \times 10^{-45}$), small versus large companies ($p = 1.589 \times 10^{-27}$), and medium versus large companies ($p = 4.864 \times 10^{-8}$). These findings indicate that salary distributions differ across all three company size groups. Therefore, the observed differences are not limited to a single pair of groups, providing further evidence of an association between company size and salary distribution. In addition, a violin graph was plotted to better visualize the salary behavior across sizes, and later another was plotted, allowing it to be broken down by flexible and non-flexible work modes.



Graphic 2: Overall median salary distribution and split view by flexible and non-flexible work modes.

The visualization presents two different perspectives on salary distribution. When analyzed alone, the median salary shows an almost linear progression, with salary increasing proportionally with company size, although a slight difference between medium and large companies is observed. Afterwards, the split distribution's view reveals a strong influence on lower median salaries through flexible positions within both small and large companies. Conversely, mid-sized companies maintain a much tighter salary grouping regardless of flexibility. Building on this observation, the next analysis explores the correlation between work mode and company size to evaluate a potential salary distribution strategy associated with the work arrangement.

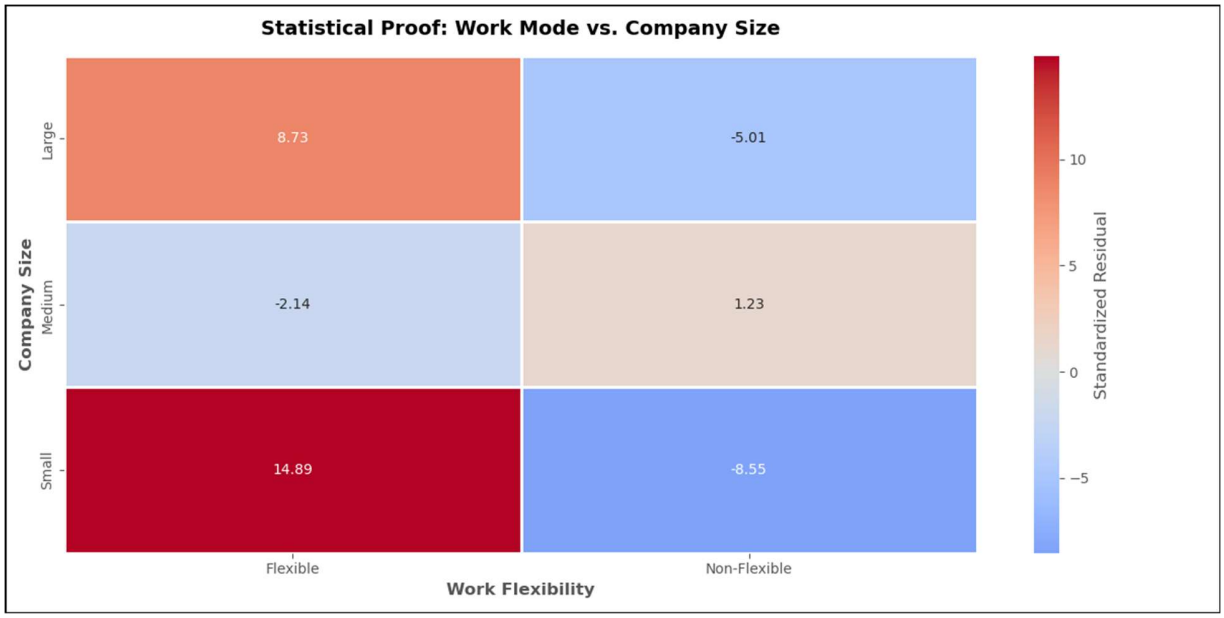
Experiment 3: Are different company sizes more related to flexibility work culture?

To evaluate potential salary distribution strategies associated with work mode, the correlation between company size and work arrangements was measured. The results were merged with those observed in previous experiments to determine possible uses for those features within companies. The test definitions are the following:

- **Null Hypothesis (H_0):** Company size and work mode are completely independent variables
- **Alternative Hypothesis (H_1):** There is a statistically significant association between company sizes and work mode

- **Alpha (α):** 0.05

Once it had two categorical variables, a Chi-Square test was conducted to measure their association. The present results ($p = 4.958 \times 10^{-88}$) indicate a statistically significant association between company sizes and flexible and non-flexible work modes. Subsequently, a heatmap of residuals was plotted to visualize deviations between observed and expected counts under the assumption of independence. The plotted graphic illustrates these discrepancies, highlighting how the distribution of work modes varies across company sizes and identifying the combinations that most contribute to the observed association. Small companies showed the largest distance from the expected value in both work modes, with a positive residual (+14.89) for flexible positions and a negative residual (-8.55) for non-flexible positions, indicating a greater adoption of flexible work than expected. Large companies exhibited a similar behavior with a positive residual (+8.73) for flexible positions and a negative residual (-5.01) for non-flexible positions. Conversely, medium companies displayed smaller deviations, with a negative residual (-2.14) for flexible positions and a positive residual (+1.23) for non-flexible positions. Overall, these results highlight that the observed distribution of work modes differs across company sizes, consistent with the association identified by the Chi-Square test.



Graphic 3: Correlation between work mode and company size.

In addition, the proportion between company sizes and flexible and non-flexible work was displayed through a distribution table, accounting for their totals within each group, allowing a complementary perspective alongside heatmap visualization of the association between the measured data, so that the visualized patterns can be connected with the underlying data and their measured distributions. The heatmap shows the association between observed and expected counts under the assumption of independence, helping to identify which combination of work mode and company size contributes most to the observed association. The distribution table, on the other hand, displays the proportion of each work arrangement within each company size, allowing a direct comparison of work-mode distributions across company sizes.

Company Size	Work Mode	Count	Total (%)	Total by Size (%)
Large	Non-Flexible	1,021	1.42%	64.29%
	Flexible	567	0.79%	35.71%
Medium	Non-Flexible	53,007	73.71%	75.61%
	Flexible	17,103	23.78%	24.39%
Small	Non-Flexible	53	0.07%	24.65%
	Flexible	162	0.23%	75.35%
Total		71,913	100.00%	

Table 2: Work mode distributions across company sizes.

These findings suggest a U-shaped flexibility pattern as company size grows, with small and large companies exhibiting higher proportions of flexible positions than medium-sized companies. One possible interpretation is that small companies, which are located at the left peak, might adopt flexible work modes as an alternative employment incentive since they are limited in matching larger companies' salaries. Medium-sized companies, in turn, might face a potential structural disadvantage, in which they are too big to operate as agile as small companies and too small to operate with the infrastructure of large companies. In addition, at the right peak, large companies have available a superior budget and infrastructure that may easily support globally distributed remote teams.

Conclusion

Although there is no significant pay gap penalty between the two extremes of flexible and non-flexible work arrangements, a different conclusion emerges when other variables are considered for analysis. Company size is an important factor that influences the salary range and

the work mode, so that combination is designed to form a U-shape, where small and large companies represent the flexibility peaks. It might be argued that there are different interpretations of what motivates the selection of work mode arrangement strategies across both sizes. A potential explanation is that smaller companies offer flexible positions as a compensation to compete with larger companies' salaries; otherwise, large companies' behavior reflects their superior budget and infrastructure. Medium companies, located at the bottom of U-Shaped, on the other hand, might face a situation where they are too big to operate agilely as a small company, and too small to operate with a large infrastructure company, which leads them to work on traditional on-site operations.